

CS103: Computer Networks & Internet Protocols

Chapter 1: OSI 7-Layer Architecture

The Open Systems Interconnection (OSI) model standardizes network communication functions across 7 layers:

Layer 7 - Application: User interface services (HTTP, FTP, SMTP, DNS).

Layer 6 - Presentation: Data formatting, encryption, and compression (TLS, JPEG, ASCII).

Layer 5 - Session: Manages sessions between applications.

Layer 4 - Transport: End-to-end communication, segmentation, flow control (TCP, UDP).

Layer 3 - Network: Logical addressing and routing across subnets (IP, ICMP, ARP).

Layer 2 - Data Link: Framing, MAC addressing, error detection (Ethernet, Wi-Fi).

Layer 1 - Physical: Bitstream transmission over physical media (Cables, Fiber optics).

Chapter 2: Transport Layer & TCP Handshake

The Transport Layer provides process-to-process communication using port numbers.

Transmission Control Protocol (TCP) is a connection-oriented, reliable protocol providing ordered byte delivery, error checking, and flow control.

User Datagram Protocol (UDP) is a connectionless, lightweight, unreliable protocol suitable for real-time video streaming and DNS queries.

The TCP 3-Way Handshake establishes a connection between client and server:

1. Client sends SYN (Synchronize) packet with initial sequence number ISN.
2. Server responds with SYN-ACK (Synchronize-Acknowledge) packet.
3. Client sends ACK (Acknowledge) packet, establishing the TCP connection.

TCP sliding window protocol manages flow control to prevent receiver buffer overflow.

Chapter 3: Network Layer, IP Addressing & Routing

The Network layer uses IP addresses to route packets across interconnected networks.

IPv4 uses 32-bit addresses written in dotted decimal format, supporting up to 4.3 billion unique addresses. IPv6 uses 128-bit hexadecimal addresses.

Subnetting divides a larger network into smaller subnets using subnet masks and Classless Inter-Domain Routing (CIDR) notation.

Routing algorithms determine optimal paths for data packets:

Distance Vector Routing (e.g., RIP) uses Bellman-Ford algorithm based on hop counts.

Link State Routing (e.g., OSPF) uses Dijkstra's shortest path algorithm based on link costs.

Address Resolution Protocol (ARP) resolves IP addresses to physical MAC addresses.

Chapter 4: Application Layer & Web Protocols

Hypertext Transfer Protocol (HTTP) is the foundation of web data exchange operating over TCP port 80.

HTTP/1.1 introduced persistent connections (keep-alive). HTTP/2 added multiplexing over single TCP connection. HTTP/3 utilizes QUIC protocol over UDP.

HTTPS encrypts HTTP traffic using TLS/SSL cryptographic protocols over TCP port 443.

Domain Name System (DNS) resolves human-readable domain names to numerical IP addresses via recursive and authoritative name servers.

Network sockets provide endpoints for IPC over IP networks using IP address and port number.